

ActInf Textbook Group ~ Cohort 7 ~ Session 25 (Chapter 10) ~ 4/24/2025

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All right. Hello fellows. We are in by all independent accounts the last of the meetings of cohort 7 in chapter 10. We've already had some interesting sharings and consolidations. So what would be interesting in this time? How how wh where if chapter 10 is the takeoff, where do we go? Or where are we in our own chapter 10? Well, for me, for me, you say Stephen. Yeah. I was just going to say it's model building. It's applying applications is where I'm at.

Andrew.

Um well, for me, um it's just an achievement to have finished reading the book. I still have to read the appendices, but um uh upon reading the book, I feel like I have a sense of the complete picture, maybe in a murky way. Uh and then what it invites me is a couple of u things. One is to be able to tease out the flesh and the bones I guess or or to say like what is actually active inference saying and then what is kind of like additional ideas that come out from applying active inference because those additional ideas they could be brilliant or they could be just not so brilliant. Uh but that's a separate thing. So um and you were just talking about that if you just look at

the bare bones active inference that's actually a common theme across all different types of let's say um I think it was like learning you know large language models or or different types of AI approaches. Um so um what that means is that in order to really appreciate active inference it's important to look at it minimalistically like not what are all people trying to do to apply but really what is it like? It's a little bit for me like calculus like calculus is absolutely important for modern physics. Um and so you wouldn't want to make calculus more physical than it is. it's just a mathematical um way of thinking and then you wouldn't maybe want to say that the calculus is a theory of everything but to be able to really appreciate so what exactly is calculus and what is it contributing you know and how does that relate to linearity let's say or not uh and things like that uh I think that's where I'm at and I think like in order to do that maybe the next step for me is just to uh compute you know so uh to learn how to uh use the Python uh modules I guess and to compute things that would be relevant for me and get some hands-on experience. I think that's probably the next step for me. And then maybe after that like then to say okay take a fresh look at let's say what are baianesian statistics and the related math and stuff and work from there. That's where I'm at. And maybe to say like uh what's

attractive for me? It does seem uh to be
um very sweet in terms of putting
perception and action in a in a unified
footing. But that's a very particular
case where you're assuming that you're
looking at the world with a generative
model and then from the point of view of
that model you know you have this active
look at perception and so you have this
active look at action let's say and then
you have this ability to quickly switch
from one to the other based on I guess
free energy um the the free energy uh I
guess maximizing or minimizing the free
energy

So that's all fine, but that's that's
assuming a certain, you know, outlook.
Uh, and I don't think that every outlook
is always based on a generative model.

Um, so there's alternative outlooks,
let's say. But,

um, so just to kind of tease that out,
what what what kind of alternative
outlooks other than looking at systems
in terms of like their input and their
output?

Well, I just don't think like I don't
think necessarily that the physical
world is based on a generative model,
you know, because a generative model is
kind of like a it's a symbolic
assumption that like so in the way I
would think about like um there's
knowing versus not knowing, you know, so
like there's you know maybe values and
variables and so um the generative model
presumes that you have this world of
variables of not knowing of symbolic

language

um But I think that something like a neural network um doesn't have that. I think it's it's not about u having variables and slots. I don't think that that's a good way to uh think of a neural network uh working or or just all different kinds of wiring working. Um there's there's a and so I think more like a generative model is working on top of other things but not it wouldn't make sense I think to have a like so the notion that everything is a Markov blanket. I just don't think that that makes sense. You can look at everything as if it's a Markov blanket, but that's very peculiar that's that's a peculiar maybe important way of looking at things, but that doesn't uh that's not the only way that things would function. And I don't think physics function.

Well, I mean it returns to kind of map territory there because when you said that the neural network isn't just parameters and slots, I mean on the first level, it literally is. That is what the parameters of the neural network is. They literally are parameters and slots. And then even abstracting if we couldn't look at those parameters in the slots, there's still um an input output relationship which is modeled with an interface.

Mhm. That's kind of the the the the surface like the sort of cloud of unknowing edge of our um doors of perception that any modeler finds themselves in which is the situation

described by inferring latent states from observations and then the special case of action is not acting. So the passive case we get for free and that's why we seek to study the possible joint um co-action of perception and motor movement for example. Well, I think maybe there's a two ways. See, I think it's more about I think your points are important and and I guess they're probably correct. U but I think it's maybe then about interpretation like even with a neural network and I very know very little bit little bit about them but I think that there's two different ways to interpret them. like you can interpret them in terms of like a hard wiring where like you're just getting signals that are coming together and they're you know at any inst at any instance there's no notion of variable I mean it's just what it is it gives you what it gives you now then you can say okay but let's make it and maybe that's the point of the neuron nature is that let's make it more flexible where it could be rewired right it doesn't have to be just wired once right it could be rewired so the notion that this is plastic and that this is rewirable and that these are slots, let's say, right? Like that's almost like a different interpretation of what's going on. It's saying, well, there's a different way of looking at the neural network where it's saying this is all plastic. These are all arbitrary. You see, and I think the

fact is that those both those interpretations are important. And I my claim would be I think that both those interpretations are communicating with each other and there's a gap between those two communications. And so so that's the gap in the action perception model like the action the I mean the I'm sorry yeah well the organism and the environment in the environment you have one hardwired reality and then in the organism you have this idea that oh we can be flexible you know like we can be flexible but whereas the environment is considered it is what it is so I think that Um

I I I think I see you on that view. Um and we sometimes are thinking about that like uh it's a static room and the agentic organism can navigate do the change.

But the whole point is it's joint system modeling and it could be a super dynamic room with a metronome in it or it could just be two metronomes or it could be two really dynamic things. It's like so kind of making a qualitative assumption that that the environment or the organism has any given structure or that that one is larger. It it they're just convenient. It's like here and there like there could be here and and here could be there but well or which is the context and which is the being you know I think like that's the kind of question and so the thing I'm researching and you know a bit about this I call it you know it's predis it turns out that you have

this oscillation between well who's the being and who's the context and so in quantum symmetries it's kind of you talk about the context of the context which could be the being, you know, so it's kind of like a you have a vector and you have a what's called the dual space. These are like functionals on the vector space like but then the dual of the dual is typically the vector where like you just it's the values I guess it could take. So, and it turns out with a quantum symmetry when you do the dual of the dual or you do the context of the context, you may get back to where you started, but you're also allowed to get back to the negative of where you started, you know? So, it's kind of like the the in this case it'd be like the data became the slots and the slots became the data or like the particles became the holes and the holes became the particles. And so and the idea is that there's this thing called B periodicity where you can cycle through these things and there's like eight different possibilities and you just get back. So the thing I realized yesterday was when you go from context and back um and you look at this whole cycle, it looks like the database update problem where like you have two copies. We're kind of like looking at things from two copies and then you want to be able them to be mismatched. Like if you want to you're working with a database and it's giving you a presentation, you want to be able to edit that, but then it needs

to be able to update its record. So there's going to be a mismatch and then there's going to be a re synchronization and a mismatch and recynchronization. So there's an oscillation between are your two different ways of looking at them kind of meshed so nicely together they're it's hard to distinguish or are you realizing no here's an instance and here's a general plastic, you know, ability. let's say and they need to be kept separate and they actually mismatch right now but then they'll rematch. Am I kind of making sense?

It seems like that it would be again another a feature of the model that would employ the methods. Yeah. So this is saying like active inference is a very nice beautiful or maybe even slick like way of talking about one aspect of this and then you have to okay so what's what's the models that you're going to think of this in terms of and so and that's what I'm talking about and and I think the idea is that uh between the organism and the environment is there a third mind that is kind of like facilitating that or not you know is it needed or not but if you have this kind of like database update problem and or you have this kind of like mismatch like or like you know I have this emotional situation where like I don't feel things are right let's say right and you're trying to model that it may suggest that there's above and beyond that these things are not self-regulating there's a regulator on

the side that's contributing to you know cybernetically managing it that's what like full-fledged consciousness would be okay this now is running up against something that I'm constantly struggling with. uh this um free energy principle is a natural uh uh uh feature of existence I feel and it's being this this framework um is being applied to our framework which our our human conceptual framework which is completely ruled by our conceptualizations I mean nothing we do lives outside of the constraint of how our minds work. Uh we think the world works the way it does because of the nature of our construction, our bodily ability to conceptualize and these concepts are constrained by the way we're constructed. uh and I run up against um uh working back and forth between these two realities. The the world is the way it is and works the way it does and we interpret it the way we do uh to the best of our ability. Um there this uh I I I posted a reference to the handbook of abduction. Uh you remember Daniel the \$700 book that Oh yes. worth worth every penny which which kind of u confronts this problem of it's basically an epistemological problem. How do we know what we know and what constrains what we can possibly know? But again, everything I'm interested in with active inference is through the filter of how

the brain works, how the mind works. And
um the I was just at a seminar
yesterday on is it possible for machine
intelligence to have meta
awareness and that's something that that
that this lab is working on right now.
Uh the PI is going to a conference in
San Diego of
um I don't know what it's called but
it's machine learning. It's some big
leagues trying to address this problem
of how can these deep learning models be
aware
of themselves to the point where they
can uh filter out hallucinations
themselves where they can know or assess
how good their reasoning is.
Um so again on that in that vein and I
thought of you when I read this paper
that I posted in the chat um modeling
meta awareness because uh I know you
were working on this three minds three
brains
issue and it explicitly addresses that
uh in a way talking about hierarchical
model of meta awareness and attention
control,
perception, attentional states, and then
meta awareness states, which I think is
kind of like what you were
describing. That's right. Okay, that's
how I'm glad I I copied the link and I'm
glad that you this. Yeah, it's funny
that that had come up in some other
who knows when chapter 10 discussion,
but I added it in those kind of it was
kind of a Hallmark paper certainly in
terms of connecting multi-scale modeling.

And then here's one from that I um put
in the notes too from a few with um Lars
Sambit Smith, first author in that 2021
paper and right in others. I mean, this
this gives some some very
interesting
angles. Um, and they do test out some
things with
LLM, but like when we kind of get
against that doors of perception, like
beyond that classic Blake quote, I think
of this this letter. Yes. As a man is,
so he sees. And it's like, okay, well
then, you know, pack it up. Nothing to
see here. It's like know everything to
see here, but then how to balance like
the world as it is, doors of perception
being cleansed, seeing the world, you
know, infinity in the flower and all of
that with the
solipsism of anthropocentrism.
It's like, well, how could we escape
anthroentism when we're our own
egocentric
navigational unit and we're seeing
things through our own perception and
cognition apparatus. So, it's like what
could be the escape from
anthropocentrism? And then this part
um you certainly mistake when you say
that the visions of fancy are not to be
found in this world. Like I see that as
the reality of cognitive science,
speculation, counterfactuals, like
expected free energy rollouts, those are
in the world. That doesn't mean that
they're justified true belief. Doesn't
mean that that the

um parameters as they're estimated to be would be any level of accuracy. But visions of fancy are found in the world and and that goes in so many different ways. I mean taking us all the way up into sort of matter first ontologies which lead us to the hard question of consciousness or mind first ontologies that that have a uh counterveailing narrative on the emergence of stability.

Uh so I have a I have a question uh of something whether it could be modeled naturally with active inference or if it's something different you know which relates to this and so because I think of active inference as kind of like the second mind type of issue uh but the first mind would be just direct experience whatever that means like the answers that are coming out unconsciously popping into my brain let's say whatever and uh so there's these things I talk about as doubts and counter question. So, um, once there I was given this question, well, how do you know you're not a robot, right? And so, the it's a difficult question because it puts everything in doubt. Like, you know, if I'm a robot, how can I trust anything I know? And so, I think that there's really only one way out of that um aside from just falling back to sleep. But, u if you really want to take that question seriously, you ask a counter question. You go, well, would it make any difference? Okay. Right. And and the the way the counter question

works, it says, look, I'm not going to rely on anything I know. So the ability just kind of like clean the slate to say, look, and then it's kind of like you tell me what a robot is, right? Like maybe I am a robot, you know, or maybe I'm not a robot, but you give me some criteria so that I could tell like, you know, do robots bleed or do they not bleed or, you know, do robots have feelings? Can robots fall in love? You know, what is it? And then maybe I'm a robot, but maybe I'm not.

But I'm not going to rely on my I mean I'm not going to I'm just going to let go of my experience and build it back up. So what that does from the first mind's point of view, it's a way of letting go of itself and saying there's something maybe more out there like to get out of your context. And so the things I was talking about like how do you go from context to context to context? This is the way of letting go of your current context and being able to get into you know without a context and into another context let's say. So I think and so there's I I did like 30 examples and there was like it ended up being like seven patterns, you know, of counter questions. So like is this do I truly like this? Well, how does it seem to me? Do I truly need this? What else should I be doing? Is this truly real? Would it make any difference? Is this truly problematic? What do I have control over? Uh is this truly reasonable? Like am I able to consider

the question uh is this uh truly wrong?
Uh is this the way things should be? um
am I truly anxious? Well, what am I
doing this about this? And then there's
an eighth one that just whenever you
want, you can say, well, what do I truly
want? You see? So, so there's this whole
apparatus of these things. And I'm just
wondering, I think this is like before
you even get to having a generative
model. This is how you start to become
letting go of yourself and starting to
operate with a generative model that's
abstract, that doesn't know anything,
saying, "I don't know anything, but what
could I do with that?" This is like the
preliminary in order to build up a but
or to to to get back to thinking in
terms of a generative model. But where
would this this type of thinking fit
with active inference? I don't know. Oh,
I think it's there's there's no goal. I
think the it's
it doesn't have a goal, right?
like what do I truly just that's
completely compatible with the the
the kind of model that uses active
inference there are no priors
and then it all your prior with
experience
well so what if you don't have priors
then what happens with active inference
Daniel you can answer that it
don't sort there's the like sort of uh
substrate fabric of reality question
like what is what is the sound of the
silent bell like a like if there
really were no prior what would there

be? So that that that sort of fades into more more axiomatic metaphysics. But let's just start with a case where there there's a given prior but of a certain kind but it's a uniform within whatever kind it is. Mhm. and then popping up a level of context is like kind of structure learning like is that prior

modifiable? Well, I think what this is maybe I didn't quite absorb what you're saying, but I think like what this is saying is like I'm going to let go of all my priors. Why don't you give me some priors? Right? like you know but see what what happens there is that instead of saying I have natural priors you know I'm wired in with certain priors like I'm saying my priors could be variable like let's go with any priors you see and that's a huge shift all of a sudden because you're in a different realm you're in moving from the first mind where there's these things that are hardwired or like prior that are hardwired to saying oh no priors could be variable you see yeah this is kind of addressed in the whole Buddhist tradition of of developing a along the spectrum of you know directed attention uh awareness to uh open attention open awareness to nonduality where there are no subject object differences which is kind of like no priors in a way. it that's like uh either there is you know there only either one awareness or no separation

between awarenesses th this is sort of
where this paper takes it with
translating yes sorts of meditational
prompts into
um natural
language and using that um in terms
making parametric modifications like
modifying the temperature value
um like in a sort of uh infinite
temperature scenario like just just you
know the ball is vibrating on the
landscape in the absolute zero context
it's going to find a local minima or
maybe even the global minima. Um, in the
infinite temperature setting, the
landscape is functionally flattened. So,
whatever structure the landscape had, as
the temperature goes to infinity, it
doesn't really matter what the
temperature was. So, there's there's
those kinds of parametric
manipulations. And then there's
like usage of natural
as I mean delivering the payload itself
within a given parametric configuration
of the
model. But I mean like this is what are
you showing? What what paper is this?
This is the
um I'll I'll I'll put it in the in the
Okay. Zoom chat as well. Like this is
just from a few days ago. I mean um
obviously with with Lars with with uh
Chandaria. Yeah. Yeah. Yeah. They're
they're they're kind of going there and
uh I think it's inevitable. I think that
all roads lead there. Uh you look at the
development of for example

um learning algorithms um machine
learning algorithms uh or you know there
are machine learning uh architectures of
the uh cognitive architectures uh
different ways of approaching how to
model cognition. lots of different
models and over the years they basically
all have been converging into the
same

general architecture as as they modify
themselves. And as I say, there's
probably an inevitable convergence of
all these different approaches because
it's all trying to approach the same
thing, the same truth, I guess, is what
I would say.

Yeah, I mean there
there's there there's so many aspects on
this and uh like I think of mind and
super mind orbindo and many different
wisdom cognitive methods and then it's
like and and also there's another you
know from another vision or another
perspective It's
purely quantitative structured
formal accounting
system for for cognitive things. Yeah.
It's like it's that's the deflationary
view is sort of it's just statistics.
Now Blake might call that Newton's
single vision with like the sort of
sleep of technical expertise.
Um, and and I mean it's it's embodied
within our
own ability to to see it those multiple
ways or oscillate or or hold both open.
Certainly, it's reflected in our
communities where some people at at the

syntactic or the semantic level are going to have allergies to to these kinds of prompts as well as allergies to mathematical symbols.

So then it's like so then that and then that goes to kind of like Krishna Murdy like truth is a pathless land like but how could there be a place with no paths within or out?

Yeah I I put in a link in the chat to a presentation I gave u about uh how we use variables in the mind. Uh and so like yeah if if you look there like at that any any of those like but you get the kind of idea that um when we look at a variable um there's so much hidden there because uh it's subject to interpretation like is it constant is it variable you know what does it mean for it to be fixed right like um is it just a label uh there's so much loaded in there um that if you try to tease it out you get all these you know I think like 24 different possibilities or something like that you know there's this huge uh risk richness that's being hidden, let's say. So this idea like with the priors, are those prior something that are fixed and constant and just given, you know, personal or are they become all of a sudden they become variables that you can play with, you know, that someone else could be filling in for you, right? Like, and how do you what does it mean to switch back and forth between those types of um ways of thinking? Uh that's I think it's kind of there's this chasm

that's kind of being glossed over. Um
wow we really come down to the topic of
communication the how can the
conceptualization yes but then
communication of those concepts
and I this is why we always turn to
rigorous
mathematical you know symbology because
it's the least
vague, the least uh
imprecise and yet we still have to know
when what are the appropriate symbols to
use. We can we can inappropriately apply
statistical
methods and you know uh obscure the
communication even still and yet it's I
think the best we
have or there is the extended mind idea
that I can have my conceptualization
communicate it and because you're so
similar to me in construction. You can
create your own
conceptualization that is more or less
similar to what mine is based on the
means of communication we're employing.
And because we live in a structured
environment that's the same, it
constrains our conceptualizations in a
way that are most likely going to be
most similar.
Yeah. So many ways to take this just
just trying to just trying to
triangulate it with like the abductive
work that you've shared and and what
they're what they're discussing in this
paper like we're not imposing any single
metaphysical system. However, what if
there's a single meta metaphysical

system?

Huh. Yeah. And it's like another like the the sort of show not tell realization of metacognitive science is like narrative and perspectival plurality as popping up a layer in the consideration of of all of what had been considered at a at a cognitive layer. Yes, I think this is approaching the best we can do. So in trying to speak that language but um and I have I think you're all a little familiar with this what I call these three minds like uh so I would think like as a generative model as a type of metaphysics let's say or conceptual system and then the point being for a for a conceptual system to really function well and do its job there needs to be a tiny uh and very uh very clear it's it well there just seems to be certain things that that explain that it's or that maybe um amount to the fact that it's functioning, it's able to function. And those are the things that's able to be able to swap back and forth with the things it's trying to model. Like you know that it's if it's if it's a good model, you can invert it and you can the context can be understood as the as the reality and the reality the context and flip back and forth and all these symmetries. And there's very few symmetries you have to worry about. Like there's a there's but there's so many

there's like eight. And if you can do that, you have a metaphysics of the metaphysics. And if you can't do that, you simply don't have a function. You know, you have a you have a generative model that's just not going to function um very well. It's just going to fall apart. So um so the fact that the generative model fits up with the things it's modeling is what the third mind is all about and the symmetries that need to be satisfied for that to happen.

There's which are like updating symmetries like the way a database would function. you know, how do you how does it update itself?

So interesting. I I I I totally agree with what you're saying. Uh I guess and I

um it it as a side tangent, I'm always amazed at why distillations of truths always come down to three parts. And I think that's got to be a natural principle. Well, sorry, it's not a natural principle of the world. It's a natural principle of our conceptualizing mind. There's this, there's that, and then there's the thing that's neither this nor that. You know, that's three parts. Yeah. You have you come up with the three minds. That's the most efficient way to communicate what you're trying to communicate. That's three parts. I I just find that so interesting. It always it's it's just a recurring theme in communicating features of reality.

Where I see that in terms of the blanket

is first there's agent, environment, and blanket. That's the three that's the threesome or the three-fold partition. Well, first there's there's this and that. That's the twofold. Agent, environment just with the arrows connecting them. Then a blanket is interpolated. And then the blanket is nuanced into incoming and outgoing. And so the the the two, the three and the four all represent this sort of canonical um and and also those two three and four are part of a conceptual unity which is the unity of that scenario. So it's like 1 2 3 4 as one two three and four and like that I mean it it it can go so many other ways with the with the numbers in the geometry. Yes, but the low integer numbers are awesome. Yeah, I I of course I have my own epistemology used with this tripart uh distillation that taken as a whole makes a fourth which is echoed in what you just showed on the screen. It's just but what we always have to remember is this isn't a natural law of the world of the universe all of these you know low ordered uh descriptions uh numerical descriptions it's not a natural law of the universe it's a natural law of the way our minds work well and possibly the way generative models work. So exactly well our mind is so you know so we we we'll know best about our own minds let's say

but but um but then we can we can maybe
see it in other things well we're going
to create these generative models
according to our conceptualizations
right they will work that way too yeah
and then if if they're useful in
modeling different things as sometimes
they do seem to be then that kind of
gives them that kind of gives some
extension to you know, but but I think
maybe to go beyond what Daniel was
saying, uh, like with this three minds
approach, I think Daniel, I think like
you're you're explaining like how it
looks for the second mind. And so like
that would be like
uh like the zeroth mind would be in
terms of needs and responding to needs
and you can see things like Masel's
hierarchy etc. And then but then the
first mind although
it's well it it looks at things
intellectually surprisingly because
that's what it takes to get out of
itself like it kind of to let go of
itself it becomes intellectual so to
speak but then when you have the
generative model and you have these two
minds in like with action perception
then it seems very emotional because
that's what that's what the cognition
that's what the organism reads uh from
the first mind the first mind
communicates in terms of emotion and the
second mind uh responds with cognition
that it imposes but I think in terms of
the third mind the see the the first
mind doesn't see the second mind the

second mind doesn't see the third mind
but the third mind sees them and the
third mind is this willfulness this
cognizance to say hey I have to actively
relate these things and then there may
be like a god or a default that is also
actively somehow relating that I might
just be an instrument of let's Okay. And
then how do I fit with regard to let's
say the default or the god or or
whatever like you know so that I'm in
sync or or am I separate you know that
type of question that's not maybe coming
up in these areas but you know so what's
maybe that's to say like what's the role
of a will related with God you know in
terms of trying to uh actively uh manage
um I don't think that that's maybe comes
coming up there it's it's not mentioned
in the textbook I don't think but
well you can map what you're saying on
to what's said in the but it'd be an
extension I think it says that there's
and so these symmetries and those like
maybe just to add like I talked to my
friend Thomas he's a theoretical
physicist I talked with uh but it
becomes about the conjugate transpose so
like the Hamiltonian in these and these
are like what they call a free fermionic
non-interacting system so it's like a
particle all by itself like what are the
pipe relationships it could have like
with its context. But what's curious is
that it has a Hamiltonian, which is what
you need in order to have a notion of a
coherent self that could evolve through
time because you've got to if you're

going to evolve through time, you've got to be able to maintain what you are. If you're going to go through different dynamical modes, you have to be able to summarize that, which is what the Hamiltonian does. But the Hamiltonian has this thing where the self-conjugate, it's equal to its own self-conjugate. And so the question is, well, what is that? And the idea is that it's the difference between a self as a being and a self as its context like your personality or everything that you've invested yourself in. You know, you can identify with either with the spirit you are that you inhabit something or the context you've created for yourself to inhabit and then you can flip back and forth which is what these quantum symmetries do. And if you as I'm studying it, what happens and I'll conclude but is that those two things which you're equating you can strip them apart and say oh no let's think of them separately me as a being and me as my context and then let me fit them back together but let's say swap them you know kind of swap them around or make one you know there's different ways you can try to put them together but it's about those how those two things so the generative model and the thing it's modeling can they be inverted to say oh no I'm a fact in the generative model of my environment for example.

you know this uh as you're as you're speaking and and Daniel as you're pointing out things in the textbook uh

and I you know I've seen these things
and and
had thoughts about them but I'm seeing
new things and having new thoughts about
them listening to this discussion and
and uh watching Daniel's mind at work
pointing out different things at the in
the textbook and I'm uh uh struck
by the reality of the mind being an
interaction with other minds uh just as
much as uh self you know internally
generated

and and I think what that interaction
because I've never really thought much
about that but I think these games with
other people that we play we're like uh
you know we switch from I to you and
other and all and such like who is the
being and who's the context, let's say,
or vice versa. Exactly. By
communicating, we're we're it kind of
forces us to play all these games much
more actively. Um
in reality, we can't play those games
unless we're actually in the presence of
others.

Yeah, it may be very hard to play that,
right? Well, it's it's otherwise it's
internal.

[Music]

Uh I think it certainly helps
there's there's compositionality
internally and there's compositionality
externally like when they talk about
these kind of rights of practice and
passage. It's like you write down the
model, make a simulation and and learn
from from what happens. Yes. And that's

similar to like you you expect that the restaurant's going to be tasty and maybe maybe it's it is exactly on point. Maybe it surprises you. It's like you you want a hot shower. It could be the perfect temp or you could update. It's like so and then

um you so there and then I but then uh the quote at the very beginning of 10 which is that we're we're least aware in general we are least aware of what our minds do

best and it's like it is that or how is that true? Why would that be true? Why would

awareness, you know, what is awareness and self-awareness such that it is like it's like the joke of the

party? It's the least capable, but it's the aware one within an internal

compositionality and then transposing that into the extended cognitive setting that you mentioned, Stephen. And it's like we're least aware of what what others are doing. We don't know what others are thinking. Maybe for the best. And and and certainly just looking at um and looking at a a tissue or looking at an ant colony, it's not that each cell or nestmate is solving the problem that the tissue is.

Huh? You're putting this into that context of situational awareness or or uh

not it's situated awareness. It the meta awareness is the awareness but you it

can't be a thing itself.

Yeah. Yeah. Like what is metacognition?

But

metacognition it's like it's like if you developed a software package with all these like runners, handlers, dispatchers, operators, all these kinds of apparatus, but then it's like and then what is it?

Yeah. like what's the

terminal rubber hits the road

inference. And it's like the whole the whole metacognitive apparatus is just sort of waiting for the key in the ignition for something even if it's just a binary signal coming in or even like a static signal coming in. But but the whole but it's not but the gears don't turn unless something is is there.

Mhm.

And then it's like how slash can we introduce novelty within ourself cuz obviously in a social setting amongst bodied minds we can see how novelty could be introduced. Like you could tell me about a place that you went today that that I would have otherwise not have known of.

Okay. But then there's also the compositionality revealed through the novelty of just like I don't know what word I would think of if I were to think what word should I randomly think of. It's like coffee cup or something, you know, again, oh that's cliché or whatever, but it's like but I didn't I didn't

know and there's some latent distribution. It's not that I'm going to come up with a word from a language that I've never known.

So it's not a good um random number generator for your cryptography per se, but there is some distribution it's sampled from, but what's the status of that of that fancy? Okay, real quick.

This is a uh I want to um ascribe a term to what you just portrayed. trying to think of something that you don't know that there there's a pr that's a principle of something. I don't know the term to describe that. Um

uh this is some somehow it's related the end goal of awareness, non-awareness, non-duality.

The the the the goal, the the the process of meditation and escaping

self is to get to this point of not trying but being of knowing what you don't know. just being awareness which is the state one would have to be in to be able to think of something one doesn't know.

I mean that does seem to describe what I've been trying to talk about with the third mind to say like if you let go of all the answers you know like everything you know which would be all your prejudices and if I let go let's say of all of my questions everything I I don't

know all these preconceptions this whole generative model and then I just what am I left with? Well, I'm left in this void, this abstract void of how things like that get connected, you know, like what are the possible empty chambers and the doors and those chambers and things like that like where so I'm in this kind of very abstract void that I can get to know and just just be dis disconnected I guess from everything and just be but be in full cognizance let's say um but maybe so that's that's the state I'm encouraging with an invest stigatory community. But but maybe that's the state that's the state that Buddhism encourages. I'm not uh sure.

It must be if it's if there's a long, you know, generational millennial tradition of keeping what's true and letting everything else go.

It must be the same.

And the advantage of that state is that then you can rebuild everything, you know, like just kind of like with a new with a new sense of where is this coming from? Like either you accept it or you or you've kind of reworked it or whatever. And then you can have this harmony between the generative model, the second mind and the first mind reality so to speak or unconscious or or whatever.

Well, all of all my takeaway from all of my interaction with you know variational free energy and the active inference the principle

of active inference. It's my takeaway is this is a general framework that's universally applicable and it can be applied appropriately or inappropriately depending on how it's put together what the model how the model is

constructed. But it that its power is in its generality and its disadvantage is its generality.

Uh and I think

it's Go ahead. I think it's begging the idea that it so beautifully gives us freedom. It models the freedom like do you want to perceive? do you want to uh uh act well we'll maximize you know I guess we'll minimize the free energy whatever and then but the idea is that yeah but there could be different solutions that are equally basically you know equal or equal within a very small you know error and so it begs the opportunity that there's a there's somebody who gets to choose you know or somebody who gets to say hey I'm going to make this ambiguous and then given the ambiguity then I'll choose you know what do I which way I'm going to go so because but it but it supplies is a it's a very nice and rare model of freedom uh or the or the preconditions for freedom.

Well, my again my conceptualization here is that if it depends on the application the appropriateness. So if if it's being uh applied to a natural system that's pretty straightforward in terms of physics and and physical constraints. Yeah. It

doesn't require free but yeah but if
it's being applied to uh
um a uh mind
state then
it has then it's a it's maybe harder to
model always there has to be a behavior
component I mean that's
inherent in this free energy principle
right there there has to be an
action. You know, if if the action is
just a
concept, then it's more difficult
um to construct the model properly.
But if the action is interaction with
the world, then it's a little bit easier
because we're talking about a physical
system which
is we're going to say a bounded
system. Uh may maybe I I'll give my just
concluding thought that for many systems
it does a wonderful job of like uniting
action and perception and basically
saying like things are going to be a
certain way because of how they evolve.
And so that's very physical and and
maybe even deterministic, but it opens
up the idea that uh you can game this
whole relationship where like you can
balance things out between let's say
action or perception or different
opportunities and then there could be
something that just if it's able to
alter the context, you know, to say,
okay, I'm going to reinterpret the
context and that'll force things to go
one way as opposed another way. like it
does not it it sets things up at a
tipping point where it's easy to tip you

know so it's easy to model freedom in
that way it can't you can do that you
can do that make that application
yes and and that's why I say it could be
it can be misleading if it's not
constructed properly
well so something to code yeah
so many great pieces
is so many more pieces in linear time
that that I know we could add and
discuss. H how can something mislead
which does not claim to lead? But then I
mean e every single
little reconfiguration is there when
when
we go to that sort of trilmma
tetral unity duality 1 2 3 4 1 2 3
4 mode on viewing active inference
and uh again there's there's more pieces
um but a good closure for the cohort.
So, thanks fellows for sticking out to
the um epistemically
delightful culmination and and we'll
discuss more in the coming weeks where
we go from here. Thank you Dave for
leading us on this uh journey in active
inference.
Indeed. Peace. by